

Nanoelectronics and Technology

(01.119/99.503)-2025-Course Project 1

NMOSFET I–V & Process Link: a Python Modeling & Analytics Project

Python-Based Modeling of NMOSFET I–V Characteristics Linked to Semiconductor Process Parameters

Abstract

The goal of this project is to model and analyze NMOSFET current–voltage (I–V) characteristics by integrating semiconductor process parameters such as gate oxide thickness, dielectric constant, substrate doping concentration, and interface trap charge. Using Python, students will simulate how process-induced variations influence key device metrics including oxide capacitance (C_{ox}), threshold voltage (V_T), and drain current (I_D).

The project emphasizes quantitative links between **process metrology** (e.g., Ellipsometry for oxide thickness, SIMS for doping profiles, AFM for surface roughness) and **electrical performance metrics**. Through computational experiments, the project aims to demonstrate how thin oxide layers, different high-k dielectrics, and doping levels shape device performance in nanoscale transistors.

The outcomes will include:

- Analytical computation of C_{ox} , V_T , and I_D ;
- Visualization of I_D – V_G and I_D – V_D curves;
- Sensitivity and variability analyses using Python;
- Correlation between fabrication parameters and transistor electrical behavior.

1. Objective

To quantitatively model the **I–V characteristics of an NMOSFET** by linking semiconductor **process parameters**—oxide thickness, dielectric constant, doping, and interface charge—to key **electrical parameters** such as oxide capacitance (C_{ox}), threshold voltage (V_T), and drain current (I_D).

2. Learning Outcomes

By completing this assignment, students will:

- Apply Python programming to semiconductor device modeling.
- Relate process variations (measured by metrology tools) to device behavior.
- Visualize and analyze NMOSFET I–V curves under varying process conditions.
- Perform sensitivity and Monte Carlo variability studies.

Suggested experiment matrix (numbers students can change)

Factor	Levels (examples)	Notes
t_{ox}	0.8, 1.2, 1.6, 2.4 nm	Ellipsometry link $\Rightarrow C_{ox}$
κ	3.9 (SiO ₂), 7 (Si ₃ N ₄), 20 (HfO ₂)	Material choice
N_A	1×10^{16} , $1 \times 10^{21} \text{ m}^{-3}$	SIMS link
Q_{it}	0 , 2×10^{-3} , $5 \times 10^{-3} \text{ C/m}^2$	Interface traps/AFM context
ϕ_{MS}	-0.2 , 0.0 , $+0.2 \text{ V}$	Metal/work-function selection
μ_n	$0.02\text{--}0.06 \text{ m}^2/\text{Vs}$	Roughness, phonon scatter proxy
λ	$0\text{--}0.05 \text{ V}^{-1}$	CLM for output resistance

Step 1: Model the Oxide Capacitance

- Equation: $C_{ox} = \frac{\epsilon_0 \epsilon_r}{t_{ox}}$
- Input parameters:
 - Dielectric constant (ϵ_r) for SiO₂, Si₃N₄, HfO₂
 - Oxide thickness (t_{ox}) measured by Ellipsometry or XRR
- Python task: Write a function to calculate C_{ox} for different oxides and plot C_{ox} vs. t_{ox} .

Oxide Capacitance Calculation

Task:

Write a Python function to calculate and plot oxide capacitance C_{ox} as a function of oxide thickness t_{ox} and dielectric constant ϵ_r .

Parameters:

Material	ϵ_r	Range of t_{ox} (nm)
SiO ₂	3.9	0.8–3.0
Si ₃ N ₄	7	1.0–3.0
HfO ₂	20	0.5–2.0

Expected Output:

- A plot of C_{ox} vs t_{ox} showing higher capacitance for HfO₂ compared to SiO₂.
- Units: C_{ox} in $\mu\text{F}/\text{cm}^2$.
- Observation: $C_{ox} \propto \epsilon_r / t_{ox}$.

Step 2: Compute Threshold Voltage

- Equation:

$$V_T = \phi_{MS} - 2\phi_F - \frac{Q_{it}}{C_{ox}} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right)$$

- Parameters:
 - N_A (acceptor doping) from SIMS data
 - Q_{it} (interface charge) from AFM/defect density
 - ϕ_{MS} (metal–semiconductor work function difference)
 - C_{ox} from Step 1
- Python task: Compute V_T variation with N_A , t_{ox} , and Q_{it} .
- Plot: V_T vs. N_A , V_T vs. t_{ox} .

Threshold Voltage Modeling

Task:

Calculate and plot threshold voltage V_T versus:

- Oxide thickness t_{ox} ,
- Doping concentration N_A ,
- Interface trap charge Q_{it} .

Assume:

$$\phi_{MS} = 0.1 \text{ V}, Q_{it} = 5 \times 10^{-8} \text{ C/cm}^2, N_A = 1 \times 10^{17} \text{ cm}^{-3}, T = 300 \text{ K}$$

Expected Output:

- Plots:
 - V_T vs t_{ox} : thinner oxide $\rightarrow$ smaller V_T
 - V_T vs N_A : higher doping $\rightarrow$ larger V_T
 - V_T vs Q_{it} : higher interface charge $\rightarrow$ shift in V_T
- Observation:

The model should show sensitivity of threshold voltage to both process and material parameters.

Step 3: Model NMOSFET Drain Current

- Linear region:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_G - V_T) V_D - \frac{V_D^2}{2} \right]$$

- Saturation region:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_G - V_T)^2 (1 + \lambda V_D)$$

- Python task: Generate I_D - V_G and I_D - V_D plots for multiple gate oxide materials and doping levels.

Task:

Compute I_D - V_G (transfer) and I_D - V_D (output) curves for:

- $W/L = 10$
- $\mu_n = 450 \text{ cm}^2/\text{Vs}$
- V_T from Step 2
- C_{ox} from Step 1

Expected Plots:

1. Transfer Curve: I_D vs V_G @ $V_D = 1 \text{ V}$
 - Observe exponential region for $V_G < V_T$ and quadratic rise after V_T .
2. Output Curve: I_D vs V_D for $V_G = 0.8, 1.0, 1.2 \text{ V}$
 - Linear $\rightarrow$ saturation transition visible.

Observation:

- Larger C_{ox} (from thinner or high-k oxide) yields higher I_D .
- Higher V_T (due to higher doping) suppresses current at same V_G .

Step 4: Process Parameter Sensitivity

- Vary each process parameter (t_{ox} , ϵ_r , N_A , Q_{it}) $\pm 10\%$ and evaluate changes in:
 - V_T (threshold voltage)
 - $I_{D,sat}$ (saturation current)
- **Python task:** Automate this using loops or vectorized NumPy code and visualize the sensitivity using bar or tornado charts.

Task:

Quantify the change in V_T and $I_{D,sat}$ for $\pm 10\%$ variation in:

- t_{ox}
- N_A
- Q_{it}
- ϵ_r

Expected Output:

- **Bar chart or tornado plot** showing which factor most affects V_T .
Typically, t_{ox} and N_A dominate.

Observation:

- Thicker oxide $\rightarrow$ lower C_{ox} $\rightarrow$ higher V_T .
- Higher doping $\rightarrow$ larger body factor $\rightarrow$ higher V_T .

Step 5: Monte Carlo Simulation of Variability

- Assign Gaussian variations to process inputs:
 - $t_{ox} \sim \mathcal{N}(\mu, \sigma)$
 - $N_A \sim \mathcal{N}(\mu, \sigma)$
 - $Q_{it} \sim \mathcal{N}(\mu, \sigma)$
- Generate 1000 samples to estimate yield distribution of V_T and I_D .
- **Python task:** Create histograms and compute standard deviation, mean shift, and yield loss.

Monte Carlo Simulation (Variability Study)

Task:

Perform 1000 Monte Carlo runs using Gaussian variations:

Parameter	Mean	σ (variation)
t_{ox}	1.5 nm	0.05 nm
N_A	$1 \times 10^{17} \text{ cm}^{-3}$	10%
Q_{it}	$5 \times 10^{-8} \text{ C/cm}^2$	20%

Quantify the change in V_T and $I_{D,\text{sat}}$ for $\pm 10\%$ variation in:

Expected Outputs:

- Histogram of V_T distribution
- Histogram of $I_{D,\text{sat}}$ distribution
- Compute **mean, σ , and yield** (fraction within $\pm 5\%$ of nominal spec).

Observation:

Process-induced variations cause device-level spread, reflecting real fab yield behavior.

Step 6 – Correlation with Metrology Data (Optional Advanced Task)

Task:

Load a CSV dataset (synthetic or measured):

- Use Python regression (e.g., `sklearn.linear_model.LinearRegression`) to fit measured vs modeled V_T .

Expected Output:

- Scatter plot: Measured vs Modeled V_T
- R^2 value and error metrics.

Observation:

Confirms how process metrology translates to electrical behavior.

Plot Type	Expected Trend
C_{ox} vs t_{ox}	Inversely proportional; steepest for SiO ₂
V_T vs N_A	Logarithmic increase
I_D - V_G	Quadratic dependence beyond V_T
I_D - V_D	Linear to saturation knee
Sensitivity Plot	Highest bar for t_{ox} , then N_A
Monte Carlo Histogram	Gaussian spread around nominal V_T

6. Deliverables

- **Python notebook (.ipynb)** with code, plots, and inline discussion.
- **PDF report** (2–3 pages) summarizing:
 - Key results and observations
 - Physical interpretations
 - Design implications (trade-offs in threshold tuning).

Report prompts (what students must answer)

1. Which **process knob** most strongly shifts V_T and why? Contrast t_{ox} vs N_A vs Q_{it} .
 2. For a target $I_{D,sat}$ @ $V_G = 0.8$ V, what **combination** of $(t_{ox}, \kappa, N_A, Q_{it})$ meets spec with **maximum** variability margin?
 3. How does switching from SiO_2 to HfO_2 at constant EOT change I_D and **subthreshold** V_T in your model?
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- Summarize relationships among t_{ox} , N_A , Q_{it} , ϵ_r and transistor performance.
 - Discuss trade-offs between low V_T (high drive current) and leakage.
 - Suggest process modifications to improve $I_{D,sat}$ without excessive subthreshold leakage.